Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 4

Full Marks: 30 *Deadline: 05:40 PM, 14-08-2025*

Submission Instructions:

1) Answer all the questions.
2) Write all the scripts, execute and take screenshots of the scripts, name the scripts files as <Your_Roll_No>_<Lab_Activity_No>_<Question_No>.
3) Prepare a report pdf which contains the question number and screenshot of script code and corresponding outputs. Report quality will contain **5 marks**. You should prepare a zip folder which will contain the report and all the script files.
4) Naming of the zip file should strictly follow the given format:
   <Roll_No>_<Name>_<Activity_No>
   If your name is Shareen, roll number is B24CS055 then the filename should be:
   B24CS055_Shareen_04.pdf
5) **Plagiarism is strictly prohibited.**

Objectives:

1) Learn the basics of shell scripting.
2) Apply conditional statements and loops.
3) Practice generating patterns and formatted outputs with nested loops.
4) Enhance problem-solving skills by integrating Linux commands within scripts.

Task 1. Write a Script to detect whether a number is prime or not. [5 marks]
Sample output:

```
arindam@ARINDAM:~$ ./demo.sh
Enter the number: 5
Prime number.
arindam@ARINDAM:~$
```

Task 2. Let's follow the grading policies: [10 marks]
80-100 => A,
60-79 => B,
40-59 => C,
<40 => Fail
Design a script which will take numbers as input and print the corresponding grading on the console. Numbers less than 0 and greater than 100 will be considered as invalid input and "Invalid Input!" will be printed on the console.

Sample output:

```
arindam@ARINDAM:~$ ./demo.sh
Enter the number: 20
Fail!
arindam@ARINDAM:~$ ./demo.sh
Enter the number: 91
Grade = A
arindam@ARINDAM:~$ ./demo.sh
Enter the number: -1
Invalid Input!
```

Task 3. Print the following pattern. [Strictly use loop logic for this] [10 marks]
Sample output:

```
1                   1
1 2               2 1
1 2 3           3 2 1
1 2 3 4       4 3 2 1
1 2 3 4 5 5 4 3 2 1
1 2 3 4       4 3 2 1
1 2 3           3 2 1
1 2               2 1
1                   1
```